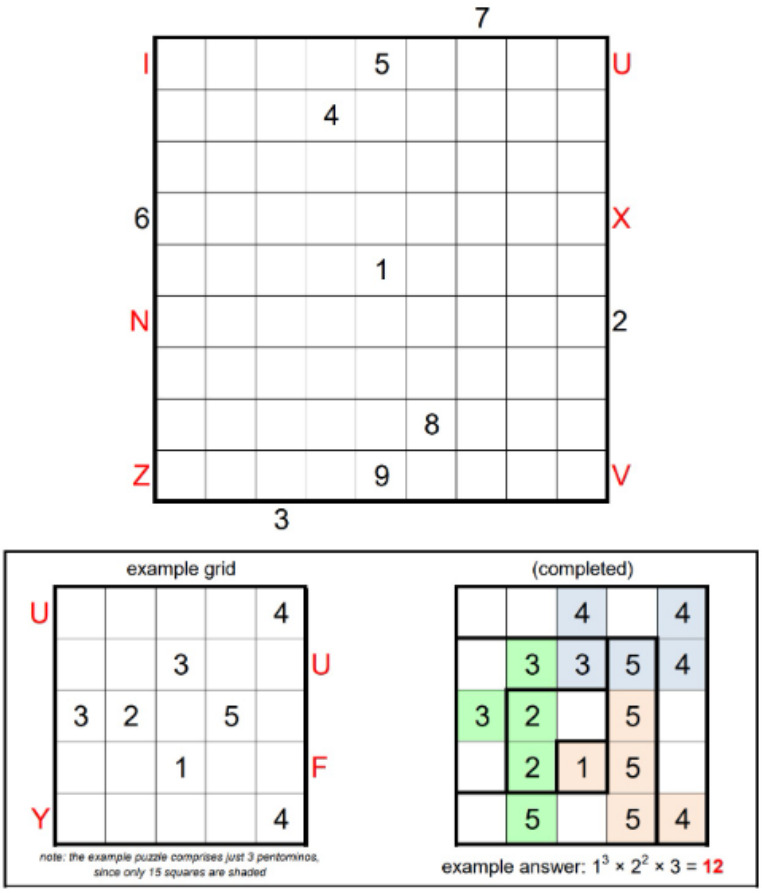
Jane Street Puzzle September 2025

Andrew Ely



The grid above can be partitioned into 9 L-shaped "hooks". The largest is 9-by-9 (contains 17 squares), the next largest is 8-by-8 (contains 15 squares), and so on. The smallest hook is just a single square. Find where the hooks are located, and place nine 9's in one of the hooks, eight 8's in another, seven 7's in another, and so on.

The filled squares must form a connected region. (Squares are "connected" if they are orthogonally adjacent.) Furthermore, every 2-by-2 region must contain at least one unfilled square.

The set of filled squares must be decomposable into **9 distinct pentominos**, with no repeated shapes (including reflection or rotation). Finally, the sum of the of the values in each pentomino must be a multiple of 5.

A value *outside* the grid denotes the either first number or pentomino[1] (from the decomposition) seen when looking into that row or column.

The answer to this puzzle is the product of the areas of the connected groups of empty squares in the completed grid.

1. Using the canonical pentomino naming scheme – F, I, L, N, P, T, U, V, W, X, Y, Z. ↩